Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

The Big 4: Deriving the Kinematic Equations

Two definitions go in. Four equations come out.

Where the Big 4 come from

Every equation on this sheet is built from **two definitions** you already have.

Average velocity $\quad \text{average velocity} = \dfrac{s}{t}$

and, when the acceleration is uniform, $\quad \text{average velocity} = \dfrac{u+v}{2}$

Acceleration $\quad a = \dfrac{\Delta v}{t} \quad$ where $\quad \Delta v = v - u$

s	u	v	a	t
distance	initial velocity	final velocity	acceleration	time
m	m/s	m/s	m/s^2	s

Reading the page. The physics runs down the left. The right-hand column does the **same algebra with plain letters**, and the same colour marks the same quantity. An arrow shows a piece moving into the next line. Each new equation ends in a **red box** with its number, and the **black box** names the one quantity it leaves out.

Equation 1

same algebra, plain letters

$\text{average velocity} = \boxed{\dfrac{s}{t}}$ $\qquad$ $\text{average velocity} = \boxed{\dfrac{u+v}{2}}$ $\qquad$ $a = b$ $\qquad$ $a = c$

$$\frac{s}{t} = \frac{u+v}{2} \qquad b = c$$

solve for s

$$\frac{s}{t} = \frac{1}{2}(u+v)$$

① $\boxed{s = \dfrac{1}{2}(u+v)\,t}$ $\qquad$ no a

Equation 2

$a = \dfrac{\Delta v}{t}$ $\qquad$ $\Delta v = v - u$

$\Delta v = \boxed{a\,t}$ $\qquad$ $\Delta v = \boxed{v - u}$

$a\,t = v - u$

same algebra, plain letters

$a = \dfrac{b}{c}$ $\qquad$ $b = d - e$

$b = a\,c$ $\qquad$ $b = d - e$

$a\,c = d - e$

solve for v

② $\boxed{v = u + a\,t}$

no s

Equation 3

— ① + ②, solve for s

② $v = u + a\,t$

same algebra, plain letters

$a = b$

① $s = \dfrac{1}{2}(u + v)\,t$

$s = \dfrac{1}{2}(u + u + a\,t)\,t$

$c = \dfrac{1}{2}(d + a)\,e$

$c = \dfrac{1}{2}(d + b)\,e$

solve for s

$s = \dfrac{1}{2}(2\,u + a\,t)\,t$

$s = \dfrac{1}{2}(2\,u\,t + a\,t^2)$

③ $\boxed{s = u\,t + \dfrac{1}{2}\,a\,t^2}$

no v

Equation 4 — ① + ②, solve for v

same algebra, plain letters

① $s = \frac{1}{2}(u+v)\,t$

isolate t

$(2\,s) = (u+v)\,(t)$ — $a = b\,c$

$t = \boxed{\frac{2\,s}{u+v}}$ — $c = \frac{a}{b}$

② $v = u + a\,t$ — $d = e + f\,c$

$v = u + a\left(\frac{2\,s}{u+v}\right)$ — $d = e + f\left(\frac{a}{b}\right)$

$v = u + \frac{2\,a\,s}{u+v}$ — $d = e + \frac{f\,a}{b}$

$(v-u) = \frac{2\,a\,s}{(u+v)}$ — $(d-e) = \frac{f\,a}{b}$

$(u+v)\,(v-u) = 2\,a\,s$ — $b\,(d-e) = f\,a$

multiply out the brackets

$v^2 - u^2 = 2\,a\,s$ *the form on your AQA equation sheet*

$(u+v)\,(v-u)$
$= \cancel{uv} - u^2 + v^2 - \cancel{uv}$
$= v^2 - u^2$

④ $\boxed{v^2 = u^2 + 2\,a\,s}$ — no t

The Big 4 — pick the one with **no** …

	Equation	Use it when there is …
①	$s = \frac{1}{2}(u+v)\,t$	no a
②	$v = u + a\,t$	no s
③	$s = u\,t + \frac{1}{2}\,a\,t^2$	no v
④	$v^2 = u^2 + 2\,a\,s$	no t

Choosing one. Write down what you are given and what you want. The one quantity left over, neither given nor asked for, is in the right-hand column: that row is your equation. ④ is on your AQA equation sheet as $v^2 - u^2 = 2\,a\,s$; ①, ② and ③ come from the same two definitions, so they are yours to use too.